

```
In [1]: #Advanced Elements Selection from ndarray---1D
```

```
In [2]: import numpy as np
```

```
In [3]: lst=[10,20,30,40,50,60,70,80,90]
a=np.array(lst)
print(a,type(a))
```

```
[10 20 30 40 50 60 70 80 90] <class 'numpy.ndarray'>
```

```
In [5]: #Step1: Identity Indices of random Elements (Ex: 10,30 and 90)
ind=[0,2,8]
```

```
In [6]: #Step-2: Pass the Random indices values to ndarray object
a[ind]
```

```
Out[6]: array([10, 30, 90])
```

```
In [8]: #OR
a[[0,2,8]]
```

```
Out[8]: array([10, 30, 90])
```

```
In [10]: #OR
ind=[0,2,8]
x=np.array(ind)
print(x,type(x))
a[x]
```

```
[0 2 8] <class 'numpy.ndarray'>
```

```
Out[10]: array([10, 30, 90])
```

```
In [11]: #Advanced Elements Selection from ndarray---2D
```

```
In [12]: lst=[10,20,30,40,50,60,70,80,90]
a=np.array(lst)
a.shape=(3,3)
print(a,type(a))
```

```
[[10 20 30]
 [40 50 60]
 [70 80 90]] <class 'numpy.ndarray'>
```

```
In [13]: a[(0,1),(0,2)]
```

```
Out[13]: array([10, 60])
```

```
In [14]: a[(0,1,2),(0,1,2)]
```

```
Out[14]: array([10, 50, 90])
```

```
In [15]: a[(0,1,2),(2,1,0)]
```

```
Out[15]: array([30, 50, 70])
```

```
In [16]: print(a)
```

```
[[10 20 30]
 [40 50 60]
 [70 80 90]]
```

```
In [17]: a[(0,0,2,2),(1,2,0,1)]
```

```
Out[17]: array([20, 30, 70, 80])
```

```
In [18]: #Advanced Elements Selection from ndarray---nD
```

```
In [19]: lst=[10,20,30,40,50,60,70,80,90,15,25,35,65,75,85,15,55,65]
a=np.array(lst)
a.shape=(3,2,3)
print(a,type(a))
```

```
[[[10 20 30]
  [40 50 60]]

 [[70 80 90]
  [15 25 35]]

 [[65 75 85]
  [15 55 65]]] <class 'numpy.ndarray'>
```

```
In [20]: a[(0,1),(0,1),(0,1)]
```

```
Out[20]: array([10, 25])
```

```
In [21]: print(a)
```

```
[[[10 20 30]
  [40 50 60]]

 [[70 80 90]
  [15 25 35]]

 [[65 75 85]
  [15 55 65]]]
```

```
In [22]: a[(0,2,1),(0,1,1),(0,2,2)]
```

```
Out[22]: array([10, 65, 35])
```

```
In [23]: print(a)
```

```
[[[10 20 30]
   [40 50 60]]
```

```
 [[70 80 90]
   [15 25 35]]
```

```
 [[65 75 85]
   [15 55 65]]]
```

```
In [ ]:
```